1. Differences between Pay-as-you-go, Reserved Instances, and Spot VMs:

Pay-as-you-go:
You pay for the cloud resources you use without any upfront commitment. It's flexible and ideal for unpredictable workloads or short-term projects but can be costly over time.

Reserved Instances:
You commit to using a VM for a period (1 or 3 years) in exchange for a significant discount (up to 72%). Best for steady workloads or long-term projects since you lock in lower prices.

Spot VMs:
Use unused Azure capacity at discounts up to 90%. These VMs can be evicted if Azure needs the resources, so they're suitable for batch processing, stateless apps, or fault-tolerant jobs.

2. Azure Pricing Calculator:
This tool helps estimate cloud costs by allowing you to:

Select services, regions, and configurations.
Adjust instance sizes, storage, and network usage.
Preview expected monthly or yearly costs.

3. Cost estimation using the Azure Pricing Calculator:

VM type: Standard_D2s_v3
Region: East US
Duration: 24/7 for 30 days (730 hours)
Price: $0.096/hour (pay-as-you-go)

Monthly cost calculation:

$730 \times 0.096 = 70.08$730×0.096=70.08

Estimated cost: $70.08/month

1. What is Azure Cost Management + Billing?
A built-in tool to:

Track cloud spending: Monitor costs by resource group, service, or department.
Set budgets: Define spending limits and get alerts when you approach them.
Analyze cost trends: Identify patterns and unexpected spikes to optimize spending.

2. How to set up a budget alert for 80% of a $1,000/month budget:

Go to **Azure Portal** → **Cost Management + Billing** → **Budgets**.
Click **Add** and set these values:
Name: "Monthly Budget"
Amount: $1,000
Reset period: Monthly
Add an **alert condition**:
Threshold: 80%
Notification: Email, SMS, or webhook
Click **Create**.

An alert will trigger when spending hits $800.

3. What is anomaly detection in cost management?

- Anomaly detection uses **AI** to automatically **spot unexpected spending patterns** — like a sudden spike in VM usage.
- When anomalies are detected, Azure notifies you so you can fix the issue quickly (e.g., shutting down idle VMs or investigating a configuration error).
- Helps **prevent surprise bills** and **optimize cloud costs**.

1. Three key cost-saving strategies:

Right-sizing resources:
Regularly review VM sizes, storage, and databases. Avoid over-provisioning by scaling down underused resources.

Use Reserved Instances:
Commit to a 1- or 3-year plan for predictable workloads — saving up to **72%** compared to pay-as-you-go.

Implement auto-scaling:
Configure apps to scale **up or down** based on traffic. This way, you only pay for extra capacity during peak demand.

2. What is Azure Hybrid Benefit?

A program that lets you **reuse your existing Windows Server or SQL Server licenses** in Azure.

Save up to **85%** compared to standard rates.
Works best when **migrating on-prem software to the cloud**.
Helps reduce **double-paying** for licenses.

3. How auto-scaling optimizes cloud expenses (with example):

What it does: Adjusts resources automatically based on demand.
Example:
An **e-commerce app** experiences traffic spikes during a flash sale.
Auto-scaling increases the number of VMs to handle the load.
After the sale, it scales back down.

Benefit: You only pay for **extra resources when needed** — saving money during off-peak hours.